

Testing Report

Word Frequency Analyzer C# CLI

Repository:

<https://github.com/fa1dpip/word-frequency-analyzer-csharp>

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The project contains two code versions: Version 1 for basic word counting and Version 2 with configurable ending trimming using parameters N and M.

1. Objective

The application reads all .txt files from a selected directory, loads their contents, splits text by whitespace and punctuation, converts words to lowercase, counts frequencies with Dictionary<string, int>, and prints the result.

Version 2 extends this behavior: if a word length is greater than N, the last M characters are removed before the word is counted.

2. OOP Structure

Class	Responsibility
FileReader	Validates a directory and reads UTF-8 text from .txt files.
TextProcessor	Splits text into lowercase words and ignores empty entries.
WordCounter	Stores and updates word frequencies using Dictionary<string, int>.
EndingTrimmer	Version 2 class that applies the N/M ending-removal rule.
ResultPrinter	Prints sorted word-frequency results to the console.
Program	Parses CLI arguments and coordinates the workflow.

3. Test Environment

Item	Value
Operating system	Windows
Compiler	.NET Framework C# compiler 4.8, C# 5
Build output	compiled/Version1/Debug and compiled/Version2/Debug
Test command	powershell -ExecutionPolicy Bypass -File .\tests\run-tests.ps1

4. Test Cases

No.	Scenario	Input	Expected result	Status
1	Version 1 basic counting	Two small .txt files with mixed case words and punctuation	hello=2, world=1, cars=1, engine=2	Passed
2	Version 2 ending trimming	Same files, N=3, M=1	cars, car, CAR counted together as 3	Passed
3	Version 2 threshold check	Same files, N=6, M=2	No sample word longer than 6 is changed	Passed
4	Build verification	Build script compiles both versions	Two Debug folders are produced with executables and PDBs	Passed

5. Console Output Samples

Version 1:

```
hello 3
world 3
```

car 2
engine 2
cars 1

Version 2 with N=3 and M=1:

car 3
hell 3
worl 3
engin 2

6. Deliverables

- Source code in src/Version1 and src/Version2.
- deliverables/source-code.zip with the source folders, build script, tests, and README.
- Git tags version-1 and version-2 for both code versions.
- deliverables/txt-file-examples.zip with four different .txt files, each larger than 1 MB.
- deliverables/WordFrequencyAnalyzerV1_Debug.zip and deliverables/WordFrequencyAnalyzerV2_Debug.zip.
- deliverables/TestingReport.pdf.

7. Conclusion

All planned tests passed. The application meets the requirements for case-insensitive counting, punctuation handling, Dictionary-based frequency storage, and OOP separation of responsibilities.